

Hands-on Lab: Getting Started with Cognos Analytics

IBM Cognos Analytics is an AI-fueled business intelligence platform that supports the entire data analytics cycle, from discovery to operationalization. It provides users with data discovery capabilities to visually explore and interact with their data to identify the key insights for improving data driven decisions. Users can perform data discovery and then quickly assemble that information into interactive, visually appealing dashboards; all without the need of formal training.

In this lab, first you will learn how to sign up for IBM Cognos Analytics trial plan, and learn general navigation around the Cognos Analytics user interface (UI). Next, you will learn how to upload external data files to Cognos Analytics, and then learn how to start a new dashboard with templates. Lastly, you will learn how to create a simple dashboard.

Software Used in this Lab

Like the videos in the course, for the hands-on labs we will be using IBM Cognos Analytics trial version (currently limited to 90 days) as this is available at no charge.

Dataset Used in this Lab

The dataset used in this lab comes from the VM designed to showcase IBM Cognos Analytics. This dataset is published by IBM. You can download the dataset file directly from here: [CustomerLoyaltyProgram.csv](#) 

Objectives

After completing this lab, you will be able to:

- Sign up for a Cognos Analytics trial plan and navigate around the Cognos Analytics user interface.
- Upload external data files to the Cognos Analytics platform.
- Start a new dashboard with dashboard templates.
- Create a simple dashboard.

Exercise 1: Sign up for Cognos Analytics Trial Plan

In this exercise, you will learn how to sign up for an IBM Cognos Analytics trial plan.

1. Go to [Try IBM Cognos Analytics](#) .

Note: If you use your existing cloud account, you will get only 30 days trial for Cognos Analytics.

2. Fill out section **1. Account information** with your information and click **Next**. The email address you are going to use here, will be called IBMid. If you already have an IBMid, click **Log in**. Enter your IBMid and password.



On-line training - Data Visualizations - Trial

Learn the skills necessary to become a data analyst using Cognos Analytics. By registering for the IBM Data and Analyst Certificate, you have the opportunity to use the full suite of its capabilities for 90 days while you complete your course. Cognos helps you to understand your data and easily create compelling visualizations and reports.

Already have an IBM account? [Log in](#)

If you already have an IBMid, click Log in

Start your free trial

1. Account information

E-mail ⓘ
Your email address will become your IBMid, which you'll use to log into IBM.com.

██████████@gmail.com ✓

We'll send you a 7 digit code that you'll use to verify your email in step 3.

First name Last name

Sandip Saha Joy

Password

..... ✓ ⓘ

- 8 characters minimum
- One uppercase character
- One lowercase character
- One number

Country or region of residence State or province

Canada Ontario

Next

3. Fill out section **2. Additional information** with your information. In the case of the Data Center, select one which is nearest to your location. Then click **Next**.

2. Additional information

Are you a student? (optional)

☒ Yes ☐ No

Select Data Center (optional)

US East (Washington DC) ▼

Next



4. Now enter the 7 digit code you received on your email address and click **Create account**.

3. Verify email

We emailed a 7 digit code to
@gmail.com

This code will expire in 30 minutes.

Verification token

5734728

Didn't receive the email? Check your spam filter for an email from ibmacct@iam.ibm.com.

[Resend code](#)

IBM may use my contact data to keep me informed of products, services and offerings:

☒ by email

You can withdraw your marketing consent at any time by submitting an [opt-out request](#). Also you may unsubscribe from receiving marketing emails by clicking the unsubscribe link in each email.

More information on our processing can be found in the [IBM Privacy Statement](#). By submitting this form, I acknowledge that I have read and understand the IBM Privacy Statement.

I accept the product [Terms and Conditions](#) of this registration form.

Create account



5. Click **Proceed**.

About your IBMid Account Privacy ×

This notice provides information about accessing your IBMid user account (Account). If you have previously been presented with a version of this notice, please refer to “Changes since the previous version of this notice” below for information about the new updates.

Updates to the [IBM Privacy Statement](#) since this notice was originally published provide additional information about how your personal information is processed by IBM.

Changes since the previous version of this notice ✓

What data does IBM collect? ✓

Why IBM needs your data ✓

How your data was obtained ✓

How IBM uses your data ✓

How IBM protects your data ✓

How long we keep your data ✓

Your rights



I acknowledge that I understand how IBM is using my Basic Personal Data and I am at

Cancel

Proceed



6. You are now done with the sign up procedure. You will be redirected to myibm.ibm.com/dashboard/ automatically. Wait for some moment until the Coursera on-line training - Data Visualizations trial offering becomes active.

The screenshot shows the IBM My IBM dashboard. At the top, there's a yellow banner with the text "Vous vous trouvez sur le site de United States." and buttons for "Rester ici" and "Accéder au site de mon pays/ma région". Below this is the IBM logo and a search bar. The main navigation bar includes "My IBM", "Profile", and "Billing". The user's IBM ID is displayed as "IBMId: [redacted]@gmail.com".

The "Products" section is highlighted. It includes a "Trials" subsection with "1 Offering". The offering is "Coursera on-line training - Data Visualizations", which is "Activation in progress". A "View support information" button is present. To the right, a large section titled "Let IBM provide you technology and business solutions to fit your needs" contains a paragraph about IBM's offerings and a "View catalog" button. A "Need help? Get product support" link is also visible.

The footer contains links for "Contact IBM", "Privacy", "Terms of use", "Accessibility", and "Cookie preferences". A language selector shows "United States - English".

7. Click on the **Launch** button of this offering.

Products

Trials

1 Offering



**Coursera on-line training -
Data Visualizations**

Active

Expires on Jan 24, 2021

Launch**Manage**

If it has been more than 72 hours since you initiated your Cognos Trial activation, but its still showing Activation in Progress, please let us know on the forum so we can follow up with the Cognos team on your behalf.

NOTE: The trial will not be activated for learners in countries under US sanctions.

8. You have successfully launched the Cognos Analytics platform.

IBM Cognos Analytics Welcome

Home Search My content Team content Recent

Hello. Welcome to IBM Cognos Analytics.

Unearth hidden insights with a personalized analytics experience driven by AI.

Take a quick tour
Experience Cognos Analytics by using guided walkthroughs.

How-to videos
Watch short videos to learn how to use Cognos Analytics and start gaining insights into your data.

Samples
Get inspired by our samples! See what you can build with Cognos Analytics.

Accelerator catalog
Browse a growing catalog of assets to enhance your experience. Please see the Accelerator Catalog FAQ for supp...

Recent

Knowledge Center

90 days left
Upgrade
Manage my subscription

Fast track your analytics journey with the IBM Accelerator Catalog!
Analytics platforms like IBM Cloud Pak for Data and their composite products and services such as Cognos Analytics, Watson Studio, Planning Analytics and ...

IBM Cognos Analytics: overview with customer testimonials
This video provides an overview of the key features of IBM Cognos Analytics: • An interface designed...

How to create your first dashboard (11.1.0+)
This video is intended for beginners, to help them get started with authoring dashboards. In this vi...

How to create your first report (11.1.0+)
This video is intended for beginners, to help them get started with authoring reports. In this video...

How to upload your own data (11.1.x)
In this video you will learn how to upload a file, create a basic data module, and replace/append an...

How to create an exploration (11.1.x)
Welcome to our video about how to create an exploration. In this video we will show how to create a ...

Getting started with your first story

Community What's new Launch the tour

9. From now on, if you want to sign in to the Cognos Analytics platform with your IBMid, go to myibm.ibm.com. Enter your IBMid and password. Repeat step 7 and 8.

Log in to IBM

IBMid

[Forgot IBMid?](#)

@gmail.com

☒ Remember Me 

Continue



Do not have an account? [Create an IBMid](#)

Exercise 2: Navigate around Cognos Analytics UI

In this exercise, you will learn general navigation around the Cognos Analytics user interface.

1. The goal of the Cognos Analytics user interface (UI) is to provide you with a streamlined way to get started using Cognos Analytics and view content and activities pertinent to them. You will begin your general navigation here.

IBM Cognos Analytics with Watson Application Toolbar Trial Countdown 79 Search Cognos Analytics ? 1

Maintenance Thank you for your patience. At this time the email service is now restored. We apologize for any inconvenience this has caused Dismiss

Hello. Welcome to Cognos Analytics with Watson.

You can get started right away by taking a look at our introduction video, product tour and Getting Started tab.

Watch video Take a product tour

Video Demonstration

^ Quick launch

Upload data
Upload or drag and drop spreadsheets, csv files, and other data sources.

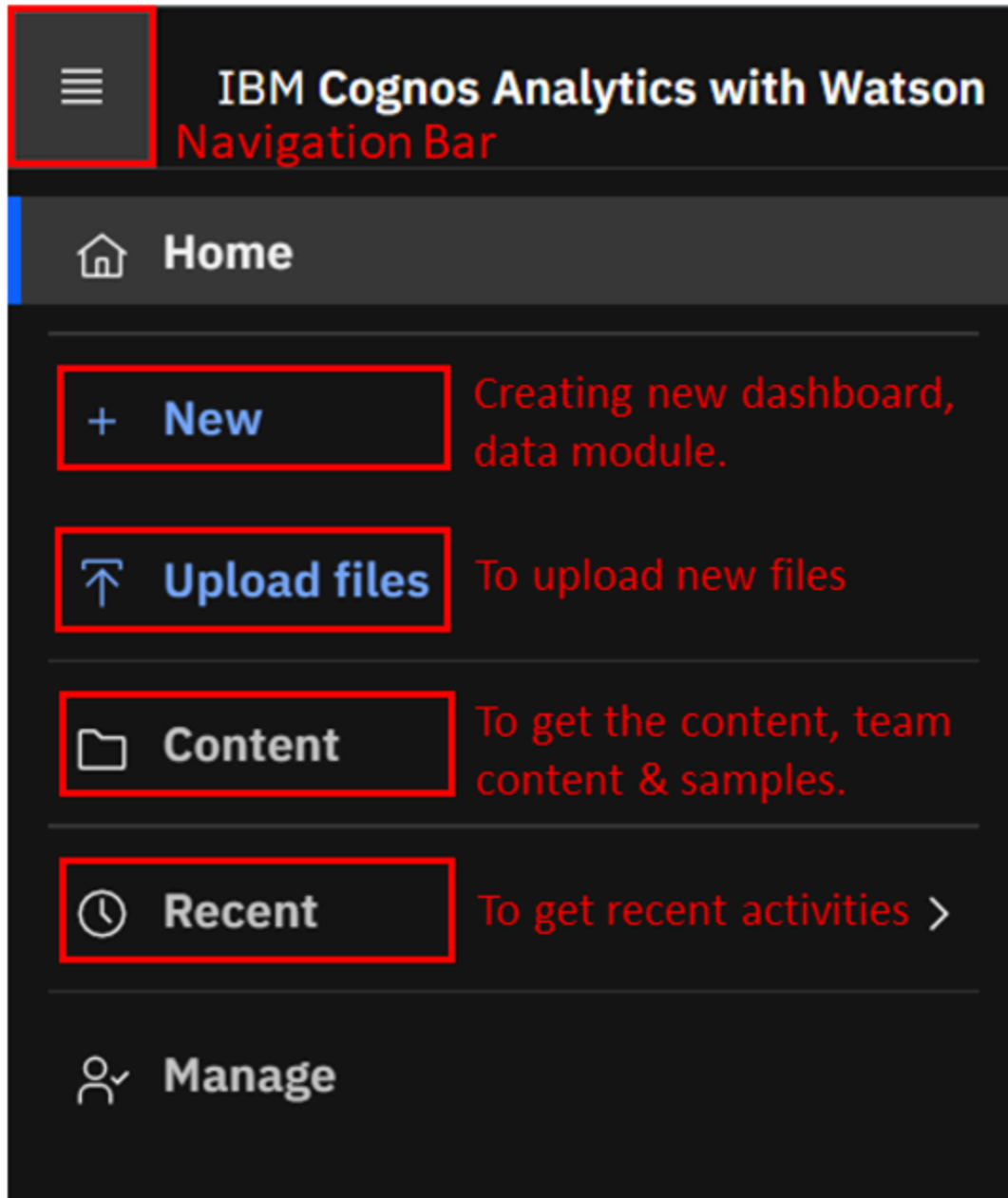
Prepare data
Use data modules to clean and connect data from multiple resources.

Exploration
Quickly find unbiased answers by identifying trends in your data with d...

Present data
Create sophisticated, multi-page, multi-query dashboards, reports, or st...

Upload Dataset

2. Click on the **Navigation Bar**, you can use the **Content** to work on different **Samples** The canvas now shows the Recently used files in the **Recent** section, if any, along with the **File drop zone** where you can easily upload your external data files.



3. Once you begin working with content, the canvas will update with your recently used items. In your Cognos Analytics instance, you may see recent content on the canvas.

Exercise 3: Create a Simple Dashboard with Cognos Analytics

In this exercise, you will learn how to upload external data files to Cognos Analytics, and then learn how to start a new dashboard with templates. Lastly, you will learn how to create a simple dashboard.

Task A: Upload External Data Files

In this task, you will learn how to upload external data files to Cognos Analytics.

1. Download the file [CustomerLoyaltyProgram.csv](#) .
2. To sign in to the Cognos Analytics platform with your IBMid, go to myibm.ibm.com/dashboard/ .
3. Enter your IBMid and password.
4. Scroll down and click **Launch**.
5. To upload a file, you may either drag and drop this file into the File Drop Zone (highlighted in the image above), or you may click **Upload files** at the bottom left corner to navigate to where the file is saved. For this lab, we will use the second option. Click **New > Upload files**.

IBM Cognos Analytics

90

Get quick answers with the Assistant

Ask the Assistant a question in your own words to uncover insights about your data.

Ask a question

Introduction to Cognos Analytics

Leverage self-service analytics to make more confident decisions.

Get started Watch video

To upload the dataset

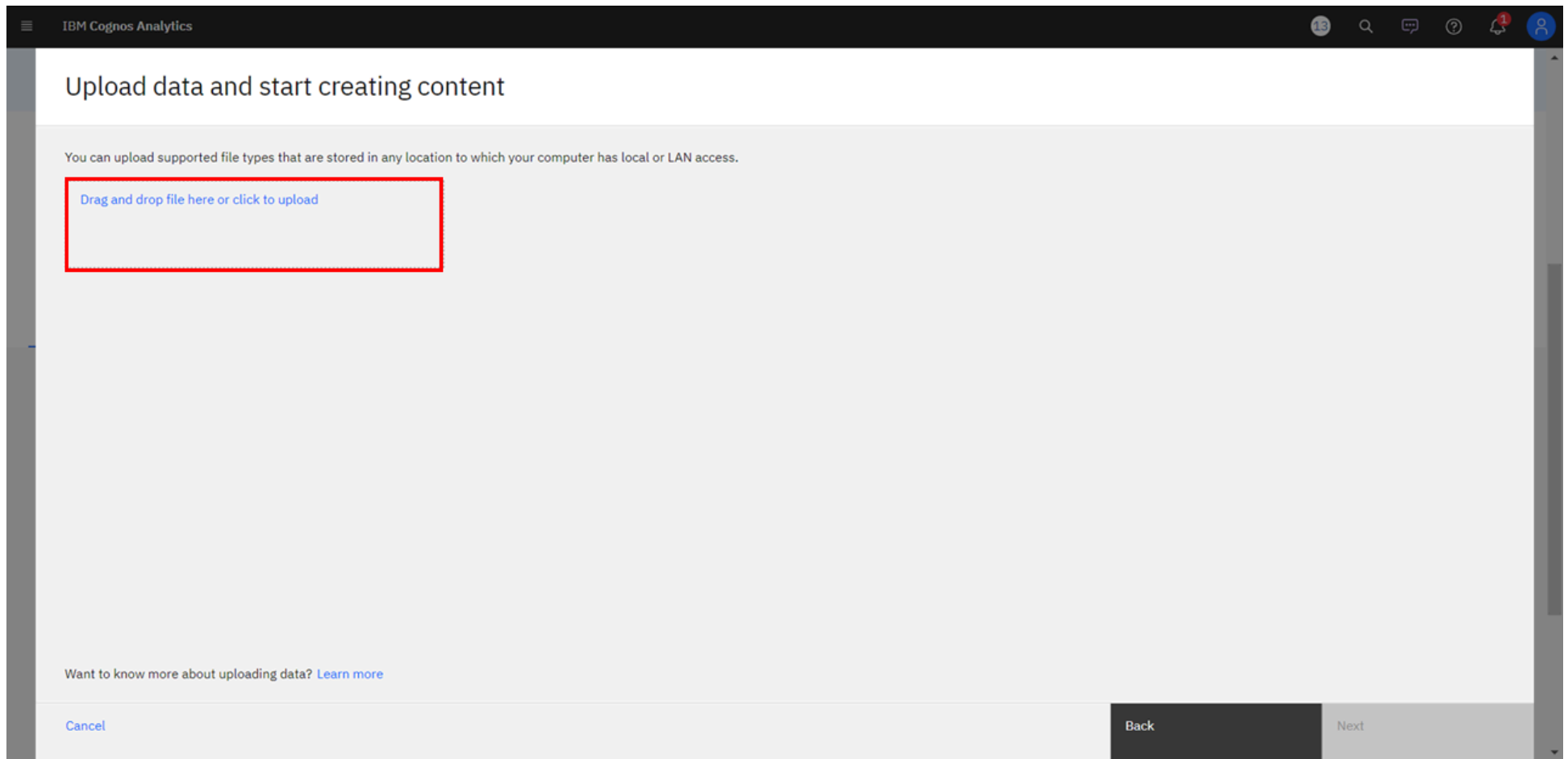
Upload data and start creating content

Upload spreadsheets, CSV files, and other types of files, and create content based on these files.

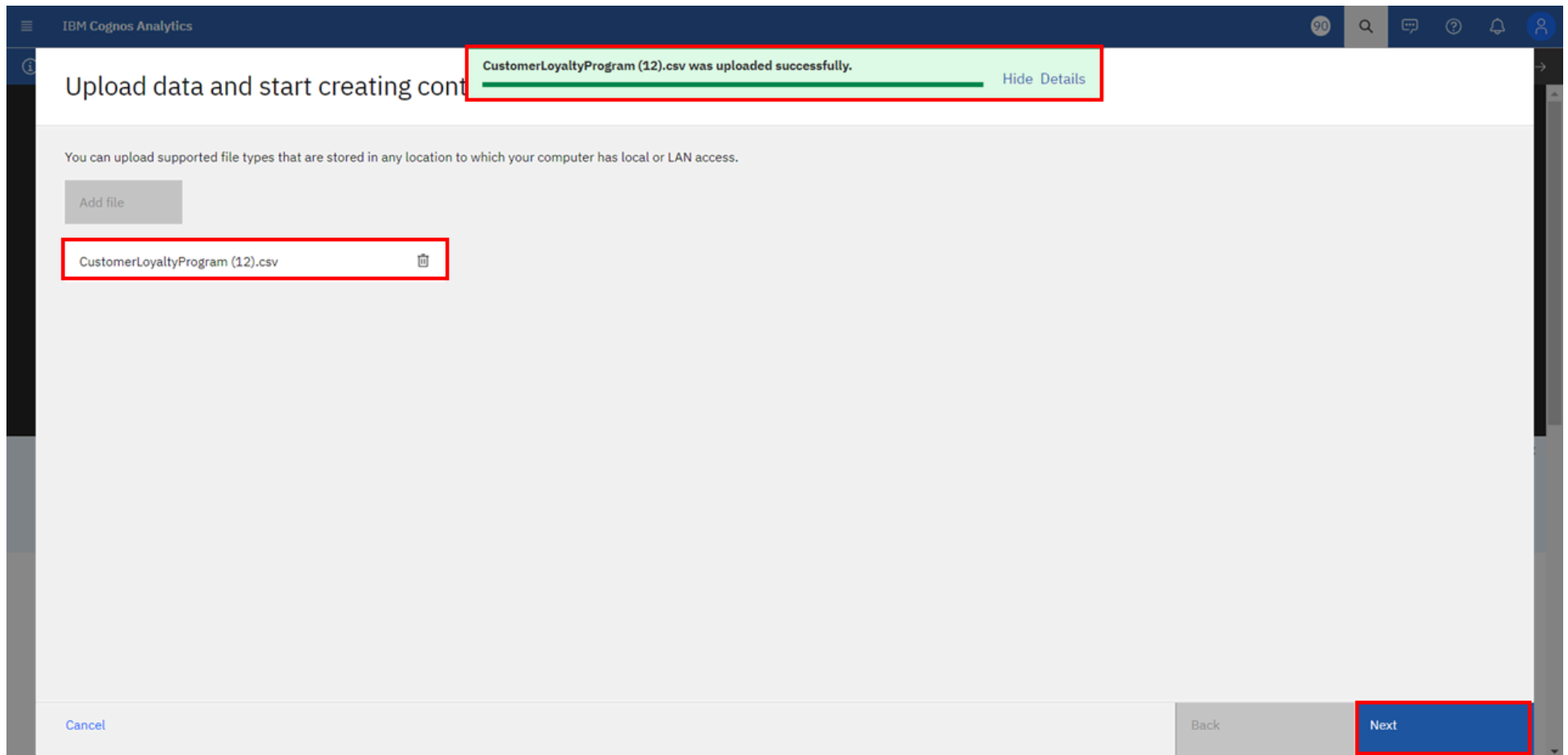
Create content from existing data

Locate data sources in the Content view, and create content based on these sources.

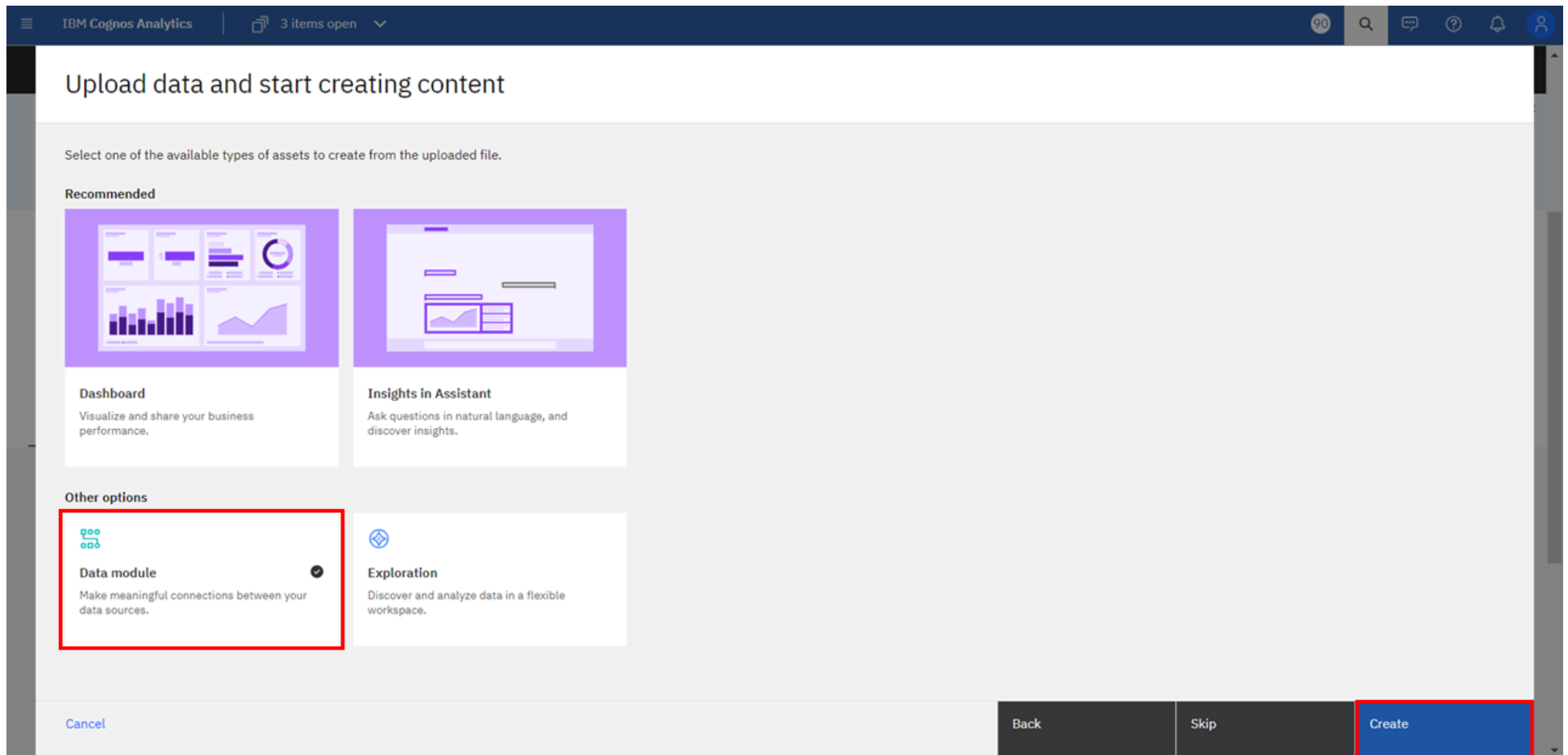
6. Click on Drag and drop file here or click to upload, a updated new file browser window will open. Navigate to where the file is saved, select the file, and click Open.*.



7. As the file uploads, status bars will be visible as the upload process reads and analyzes the data being brought in. Once it completes, the status bar will update to show the successful completion before closing.



8. Convert the **CustomerLoyaltyProgram.csv** to **Data module** by selecting the **Data module** in **Other options** and click **Create** .



Task B: Edit Data Module and view the table on IBM cognos.

In this task we will learn how to view the dataset on IBM Cognos.

1. Navigate to the **CustomerLoyaltyProgram data module**. Click **Action Menu** (represented by three dots) under it. Next click the **Edit Data Module** to create a **Data Module**.

The screenshot shows the IBM Cognos Analytics interface. At the top, there is a blue header bar with the text "IBM Cognos Analytics" and a notification banner for "Maintenance: Cognos Analytics Maintenance: 16th of September, 09:00-17:00 UTC". Below the header, there are two main sections: "Upload data and start creating content" and "Create content from existing data". In the "Recent" section, a list of items is shown, including "Customer Loyalty Program data module". A context menu is open for this item, with "Edit data module" highlighted in red. The menu options include "Create report", "Create exploration", "Create dashboard", "Create data module", "Create data set", "Share", "Copy or move to", "Add shortcut", "Edit name and description", "Properties", "Details", "Remove from recent", and "Delete".

IBM Cognos Analytics | 1 item open

Maintenance: Cognos Analytics Maintenance: 16th of September, 09:00-17:00 UTC Click on More Info to see what actions may be necessary and to subscribe to future events

Leverage self-service analytics to make more confident decisions.

Get started Watch video Product tour

Upload data and start creating content

Upload spreadsheets, CSV files, and other types of files, and create content based on these files.

Create content from existing data

Locate data sources in the Content view, and create content based on these sources.

Recent

1 item selected

Customer Loyalty Program data module

Last Accessed 9/8/2023, 5:02 AM

More + Create Remove from recent Delete Cancel

View all content

Edit data module

Create report

Create exploration

Create dashboard

Create data module

Create data set

Share

Copy or move to

Add shortcut

Edit name and description

Properties

Details

Remove from recent

Delete

2. A New Data Module will be created. Click on the Action Menu (represented by three dots) of **New data Module**. Next click the **Table** option under the **Action Menu** to launch the **Create Table** window.

The screenshot shows the IBM Cognos Analytics interface. At the top, the header bar displays 'IBM Cognos Analytics' and a dropdown menu for '* New data module'. Below the header, the main workspace is divided into three tabs: 'Grid', 'Relationships', and 'Custom tables'. On the left, a 'Data module' sidebar contains a search bar and a list of items. The 'New data module' item is highlighted with a red box. A context menu is open over this item, showing options like 'Calculation...', 'Relationship...', 'Filter...', 'Folder', 'Table using SQL...', 'Table...', 'Refresh properties...', 'Validate', and 'Properties'. The 'Table...' option is highlighted with a red box. In the center of the workspace, there is a large icon of a table with a green checkmark, labeled 'Preview data'. Below the icon, text reads: 'To preview data, select a table, a column in a table, or a folder that contains columns.'

3. Click on the **Select Tables** button in the **Create table** screen. Select the uploaded dataset **CustomerLoyaltyProgram.csv**. Finally, click the **Next** button to proceed with the creation of the table.

Create table

Create a custom table in the data module. This table is not added to your data source.

Selected tables

Find

Find

CustomerL...gram.csv



Select tables

Select and add tables to create a new table from

Select tables

☐ View of tables

Create a view of a table, select one or more package tables, or select one or more non-package tables.

☐ Shortcut to a table

Create a shortcut table, select one non-package table.

☐ Alias of a table

To create an alias table, select one non-package table.

☐ Copy of a table

To create a copy of a table, select one non-package table.

☐ Joined view

To create a join table, select one or two non-package tables.

☐ Union of tables

To create a union table, select two or more non-package tables that have the same number of columns with compatible data types.

☐ Intersect of tables

To create an intersect table, select two non-package tables that have the same number of columns with compatible data types.

☐ Except of tables

To create an except table, select two non-package tables that have the same number of columns with compatible data types.

Cancel

Next

4. Click the **Refresh** button in the **Create a view of a table** window to view the data.

Create a view of a table

New table name: CustomerLoyaltyProgram.csv - View (1)

Select items

CustomerL...ogram.csv

✓

Row Id

Loyalty#

First Name

Last Name

Customer Name

Country

Province

Data will appear here

Refresh

Invert

Clear all

(29) columns selected.

Previous

Cancel

Finish

5. Now you will be able to see view the data in the table, click on the **Finish** button.

Create a view of a table



New table name: CustomerLoyaltyProgram.csv - View (1)

Select items	Row Id	Loyalty#	First Name	Last Name	Customer Name	Country	Pr
Find	1	837810	Abigail	Goris	Abigail Goris	Canada	
CustomerL...ogram.csv ☒	2	707928	Ada	Glaude	Ada Glaude	Germany	
	3	901442	Ada	Staback	Ada Staback	Canada	
	4	237303	Adam	Boner	Adam Boner	Canada	
	5	348835	Adam	Dunnegan	Adam Dunnegan	Canada	
	6	186842	Adelaide	Drago	Adelaide Drago	Germany	
	7	605640	Adell	Maroni	Adell Maroni	United Kingdom	
	8	723924	Adolph	Prey	Adolph Prey	Germany	
	9	202299	Adria	Osterstuck	Adria Osterstuck	Canada	
	10	826271	Adriane	Dottin	Adriane Dottin	Canada	
	11	616030	Akilah	Carvalho	Akilah Carvalho	Germany	
	12	747903	Al	Johson	Al Johson	United States	

(29) columns selected.

Previous

Cancel

Finish

6. The new table view will be added in source panel now save this Data Module using the **Save** option.

The screenshot displays the IBM Cognos Analytics interface. At the top, the header shows 'IBM Cognos Analytics' and a dropdown menu for '* New data module'. On the left, a 'Data module' sidebar contains a search bar and a list of modules. The 'CustomerLoyaltyProgram.csv' module is selected, and its 'View (1)' is highlighted. The main workspace shows a diagram with two boxes: 'CustomerLoyaltyProgram.csv' and 'CustomerLoyaltyProgram.csv - View (1)', connected by a line. The 'View (1)' box is highlighted with a blue border. The right sidebar shows the 'Properties' tab.

7. In the new popup window labeled **Save as**, Give the name **Customer Loyalty Program data module** and select **My content** tab, click on the 'Save' button to proceed.

Save as

Name

Customer Loyalty Program data module

Selected destination: My content

My content

Team content

Name	Type	Last Accessed
CustomerLoyaltyProgram.csv CSV	Uploaded file	23/06/2023, 01:57
Newdash	Dashboard	18/04/2023, 06:06
popular-language (4).csv CSV	Uploaded file	26/05/2023, 04:31

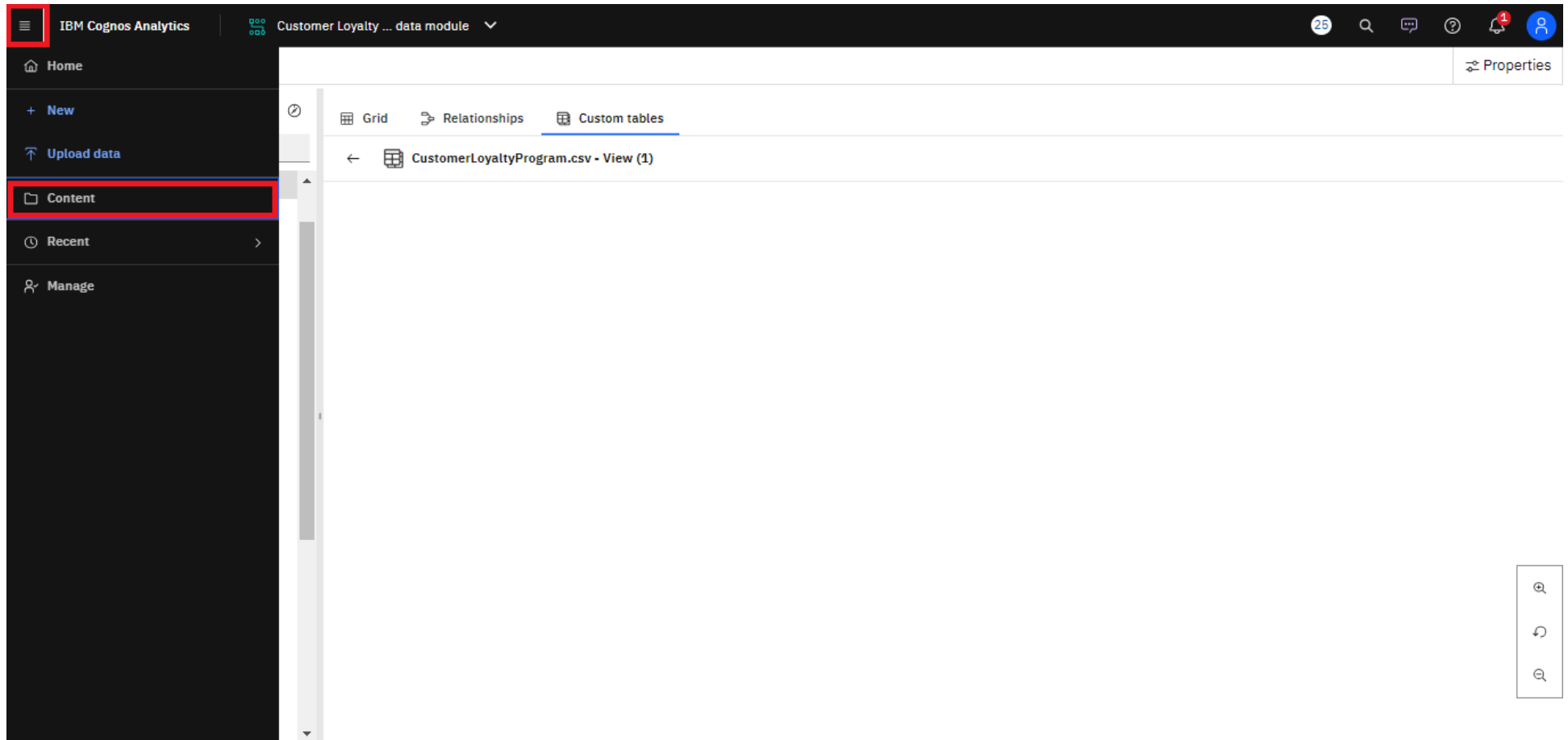
Cancel

Save

8. Next, please navigate to the menu option of IBM Cognos, which is represented by three horizontal lines. Click on the menu and select the 'Content' option from the available choices.

<https://www.coursera.org/learn/data-visualization-dashboards-excel-cognos/supplement/uWOrO/hands-on-lab-getting-started-with-cognos-analytics>

28/43



9. You will now be able to locate the **Customer Loyalty Program data module** under the **My Content** tab.

Content

My content Team content Samples

Customer Loyalty Program data module

Last Accessed 23/06/2023, 03:18

CustomerLoyaltyProgram.csv

Last Accessed 23/06/2023, 01:57 CSV

Newdash

Last Accessed 18/04/2023, 06:06

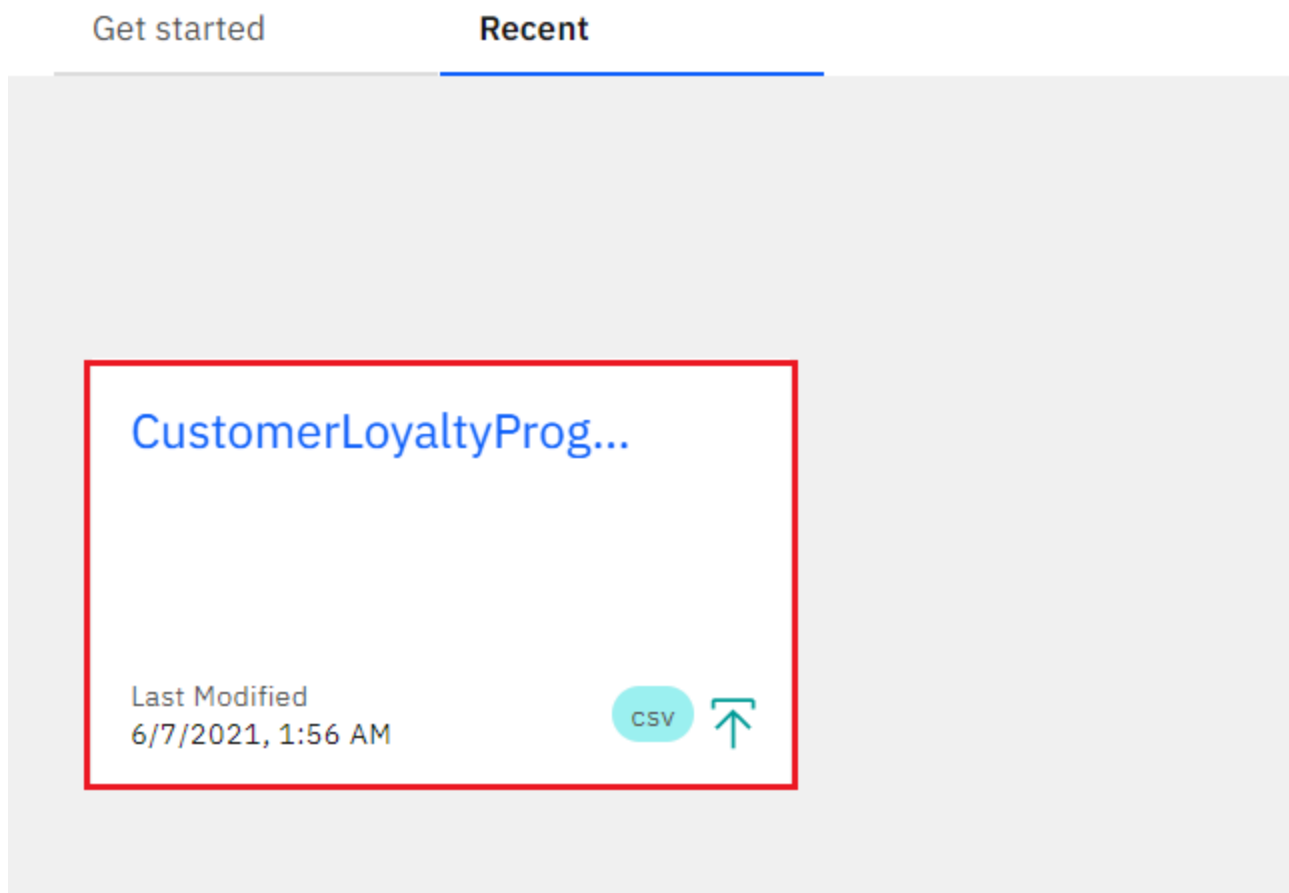
popular-language (4).csv

Last Accessed 26/05/2023, 04:31 CSV

Task C: Start a New Dashboard with Templates

In this task, you will learn how to start a new dashboard with templates.

1. Click on the uploaded data file **CustomerLoyaltyProgram.csv**.



2. The Template window will display allowing you to select the type of dashboard and the template style. Select the **tabbed dashboard style**. This will allow you to have multiple pages for your dashboards. Select the **three-panel template**. Click **Create**.

Create a dashboard

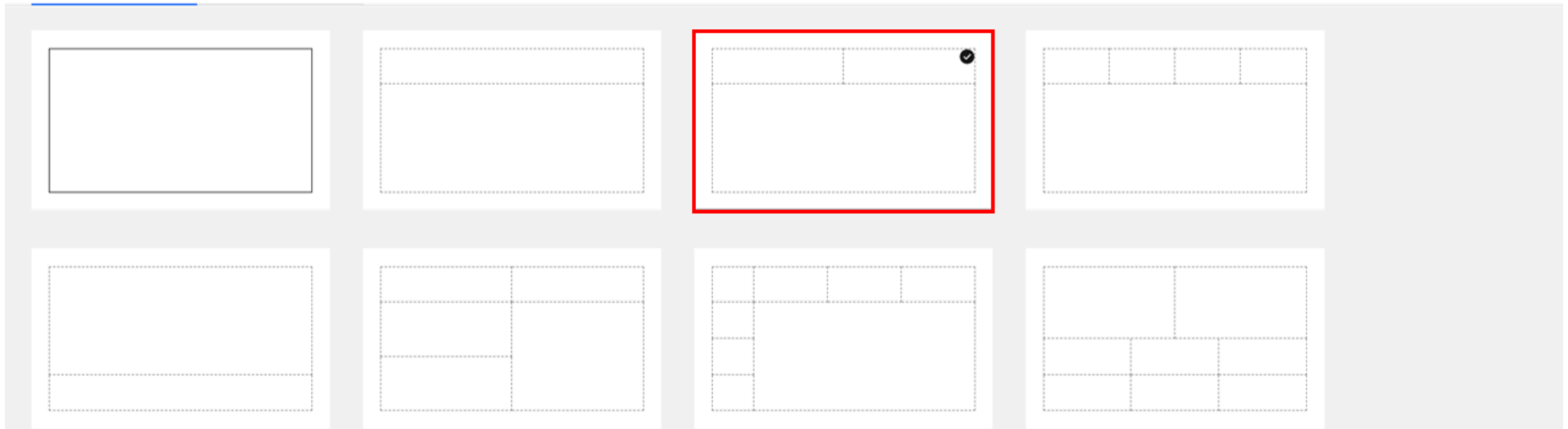
Select a template for your dashboard

Cancel

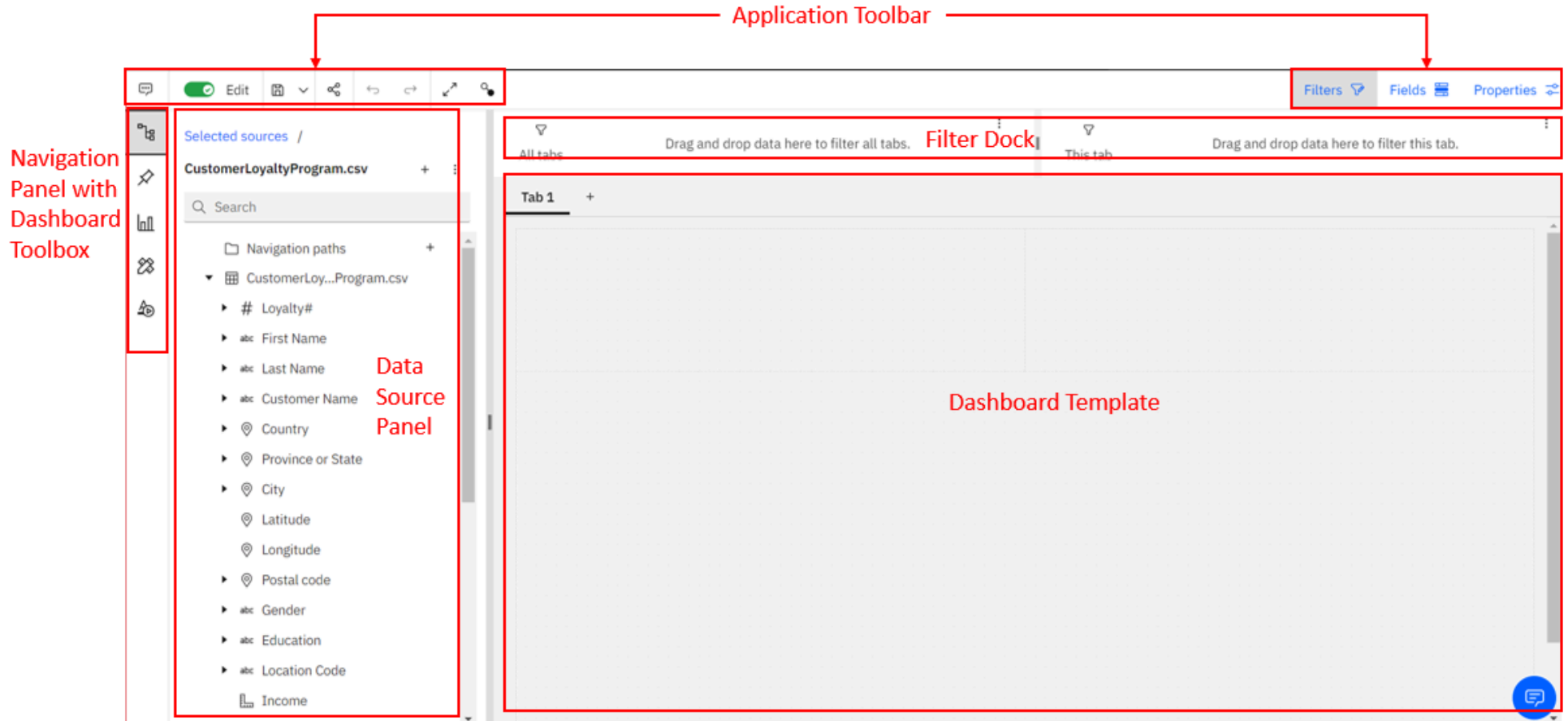
Create

Tabbed

Infographic



3. Now you have created a new dashboard using the dashboard template.



4. To save the newly created dashboard, press **CTRL+S**. Select **My content** as **Destination**. On the **Save as:** text field, write “Simple dashboard”, and click **Save**.

Save as

×

Name

New dashboard

Selected destination: My content

My content

Team content

⏏

↕

📁

⚙

Name	Type	Last Modified
📄 CustomerLoyaltyProgram.csv csv	Uploaded file	6/17/2021, 6:41 AM
📊 dash	Dashboard	6/7/2021, 2:00 AM

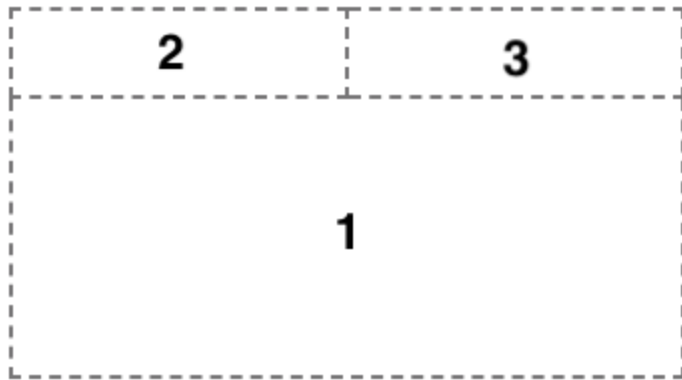
Cancel

Save

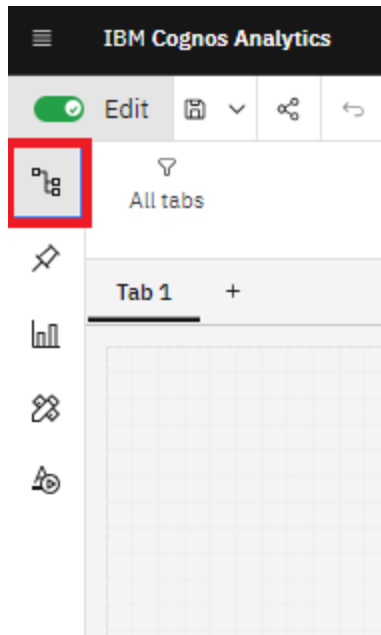
Task D: Create a Simple Dashboard

In this task, you will learn how to create a simple dashboard with Cognos Analytics.

1. As you build the dashboard, the location placement for Widgets in the dashboard template will be referenced using the following Panel numbers.



2. From the **Navigation** panel, select **Sources** to open the data source panel, if it is not already open. The **Data Source** panel displays the uploaded file ****customerLoyaltyProgram.csv** as the Selected Source.



3. From the **Data Source** panel, press **CTRL** and select **Order Year**, **Quantity Sold**, **Product Line** and drag it to the center of **Panel 1**, releasing them once you see the **drop zone turn blue**.

Search

Navigation paths

- CustomerLoyaltyProgram.csv
 - Loyalty#
 - First Name
 - Last Name
 - Customer Name
 - Country
 - Province or State
 - City
 - Latitude
 - Longitude
 - Postal code
 - Gender
 - Education
 - Location Code
 - Income
 - Marital Status
 - Order Year**
 - Quarter
 - MonthsAsMember
 - LoyaltyStatus
 - Product Line**
 - Coupon Response
 - Count
 - Quantity Sold**
 - Unit Sale Price
 - Unit Cost
 - Revenue
 - Customer Lifetime Value
 - Loyalty Count

All tabs

Drag and drop data here to filter all tabs.

Tab 1

Drop here to make a visualization

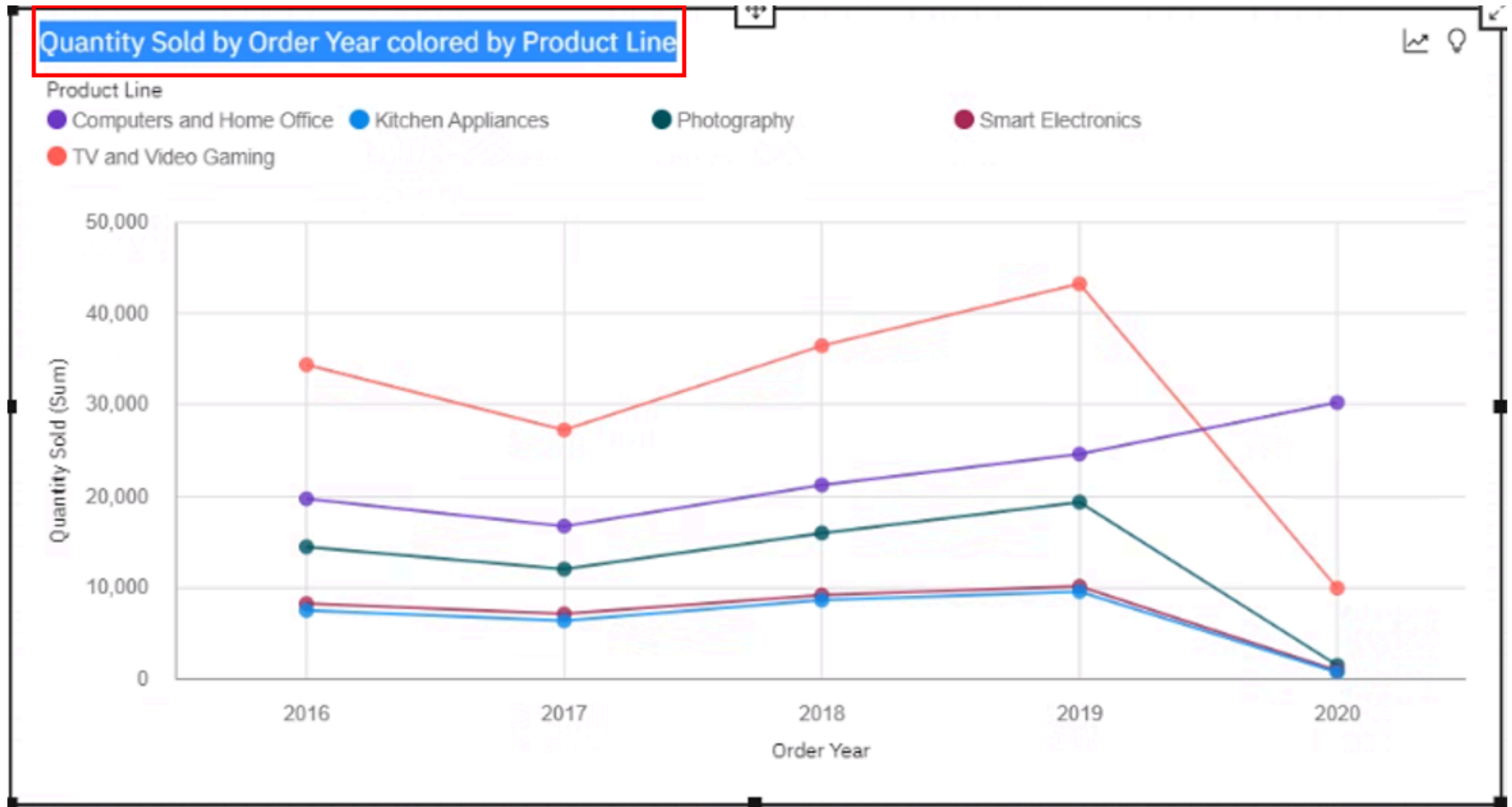
Order Year

Product Line

Quantity Sold

4. Click on the **Line chart** in panel 1 to bring it into focus and render the **on-demand toolbar**.

5. From the on-demand toolbar, select the title and enter the title “Product Line Performance by Year” to the visualization.



6. Highlight the **Title** to open the **Title on-demand toolbar**. From here, you can change the various properties on the title.

IBM Plex



A

16



B

I

U



Title On-demand Toolbar ...



All tabs

Drag and drop data here to filter all tabs.

Tab 1



Product Line Performance by Year

Product Line

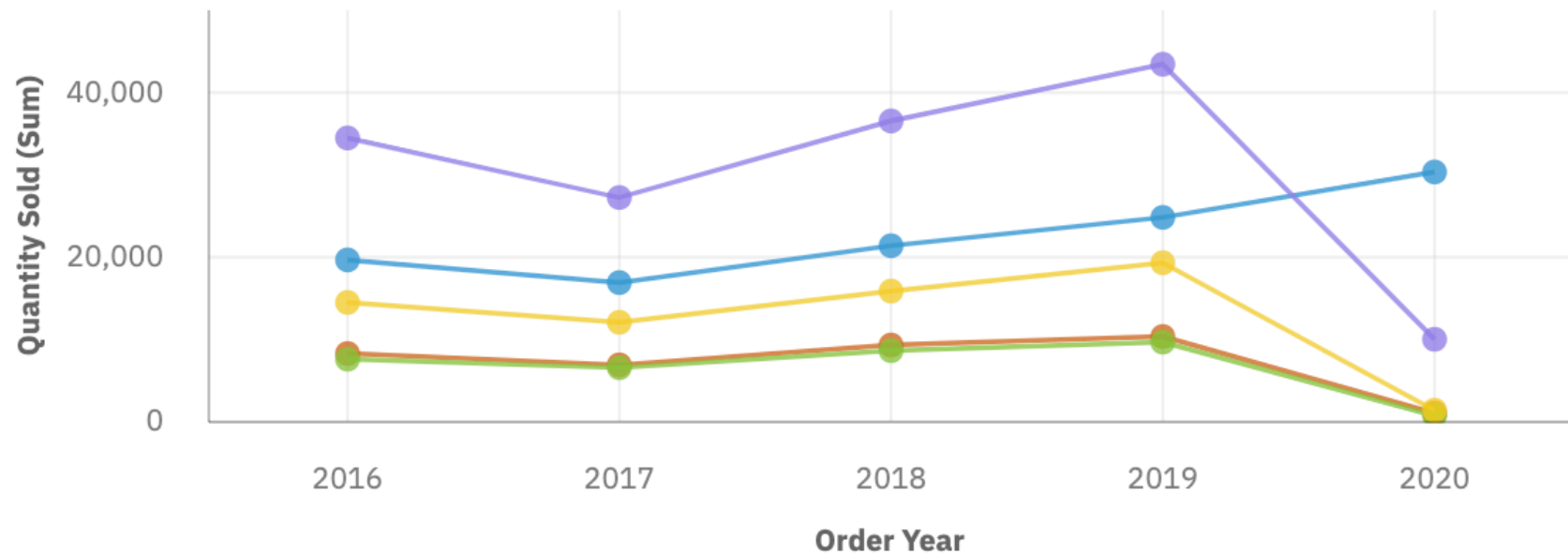
Computers and Home Office

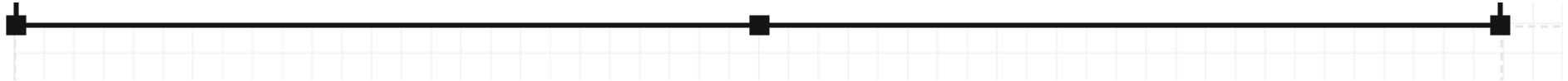
Kitchen Appliances

Photography

Smart Electronics

TV and Video Gaming





7. Click the **Color picker** icon, and change the color to **Red**, then click the font size drop-down and choose **18**.
8. From the **Data Source** panel, select **Quantity Sold** and drag it to the center of **Panel 2**, releasing it once you see the **drop zone** turn blue.



All tabs

Drag and drop data here to filter all tabs.

Tab 1



Search

Navigation paths



CustomerLoyaltyProgram.csv

> # Loyalty#

> abc First Name

> abc Last Name

> abc Customer Name

> 📍 Country

> 📍 Province or State

> 📍 City

📍 Latitude

📍 Longitude

> 📍 Postal code

> abc Gender

> abc Education

> abc Location Code

📊 Income

> abc Marital Status

> 🕒 Order Year

> 🕒 Quarter

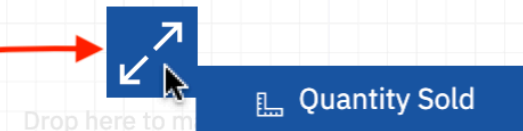
📊 MonthsAsMember

> abc LoyaltyStatus

> abc Product Line

> abc Coupon Response

📊 Count



Product Line Performance by Year

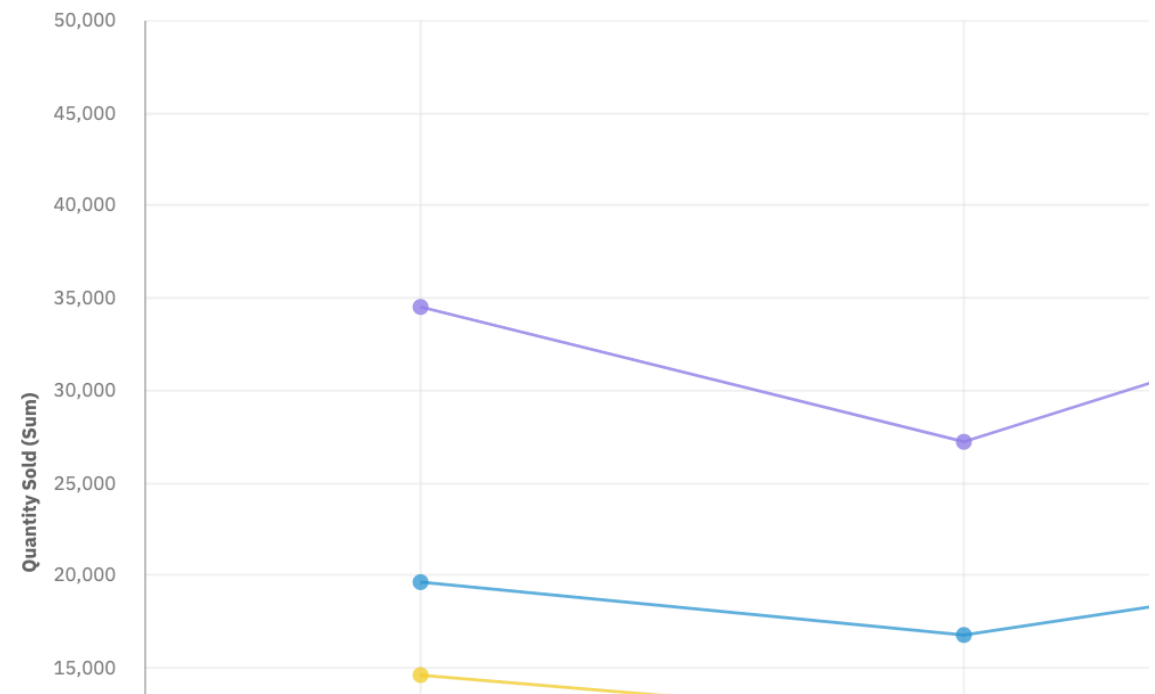
Product Line

● Computers and Home Office

● Kitchen Appliances

● Photography

● Smart Electronics

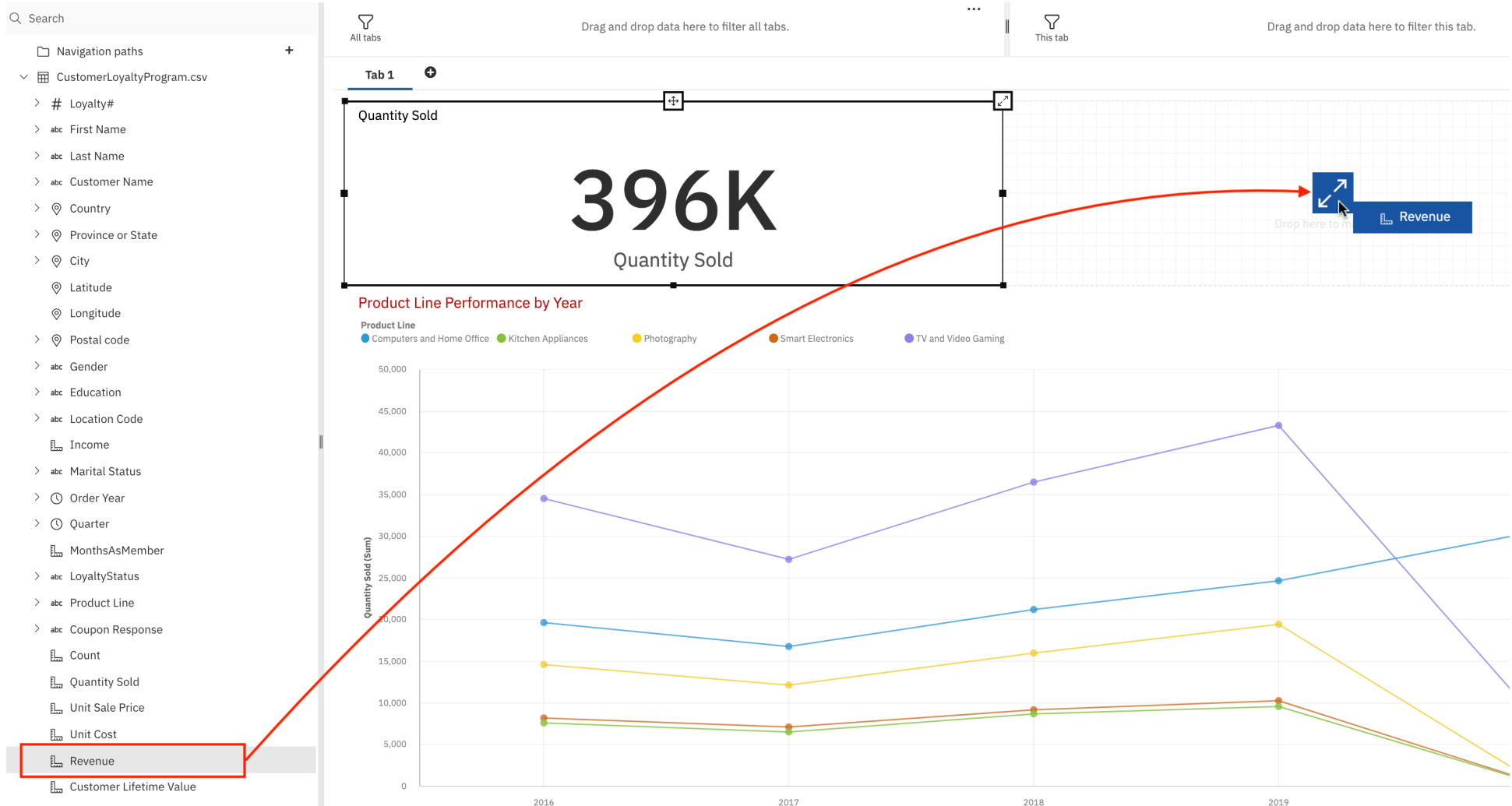


Quantity Sold

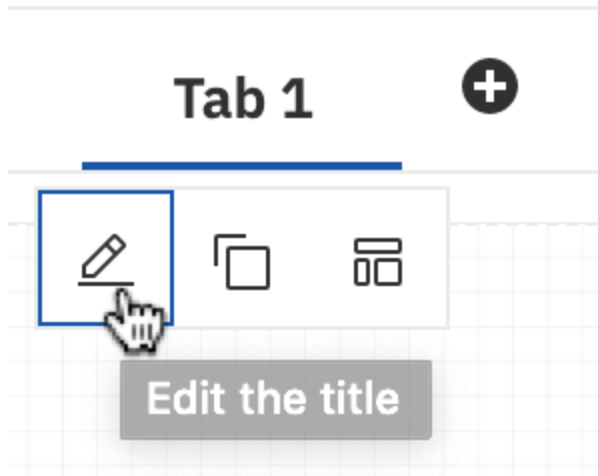
Unit Sale Price

10,000

9. From the **Data Source** panel, select **Revenue** and drag it to the center of **Panel 3**, releasing it once you see the **drop zone** turn **blue**.



10. Click on the tab name **Tab 1** to bring up the Tab on-demand toolbar. Select the **Edit** icon.



11. **Rename** the tab to “A - Product Sales” 12. To save the current work of the dashboard, press **CTRL+S**.

Finally, your dashboard "A - Product Sales" should look like below:

Quantity Sold

396K

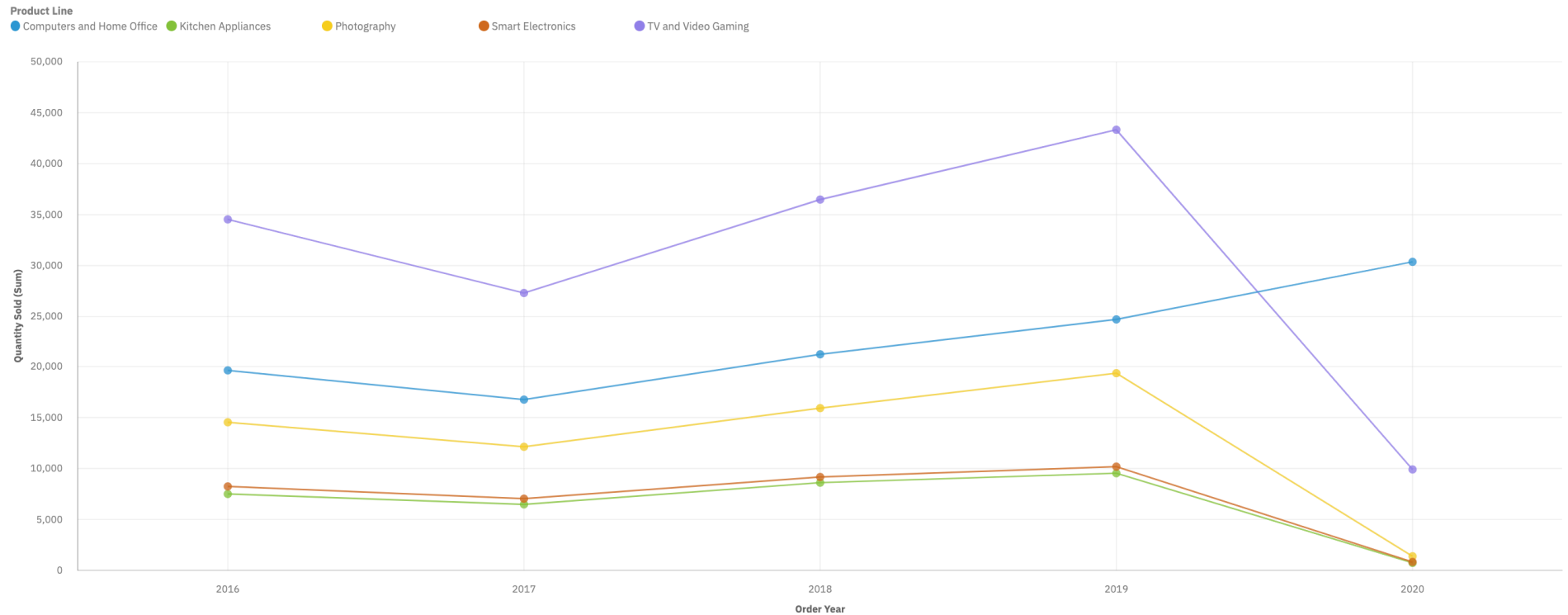
Quantity Sold

Revenue

\$176,520,879

Revenue

Product Line Performance by Year



Congratulations! You have completed Lab 4, and you are ready for the next topic.